REVIEW 3 | FINAL DEMONSTRATION

Vehicle Service Centre Management System

AutoCore VSC Website, Final Demonstration, Report & Viva-Voce (Project 10)

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Review 3 Deliverables

1

AutoCore VSC Website

Web interface for the Vehicle Service Centre database: dashboard, customers, vehicles, bookings, job cards, inventory and billing.

2

Final Demonstration

Walkthrough of the website pages, from the dashboard to billing, on the project sample data.

3

Project Report

Problem, ER and relational design, DDL, queries, implementation, results and conclusion.

4

Viva-Voce

Questions on normalization, constraints, triggers, SQL and the working website.

Project Journey: Review 1 to Review 3

1

Review 1

Design

Problem statement, ER model, relational schema and 3NF mapping

2

Review 2

Database + Terminal Prototype

3NF tables, DDL with constraints, sample data, SQL queries and a Python terminal prototype built in VS Code

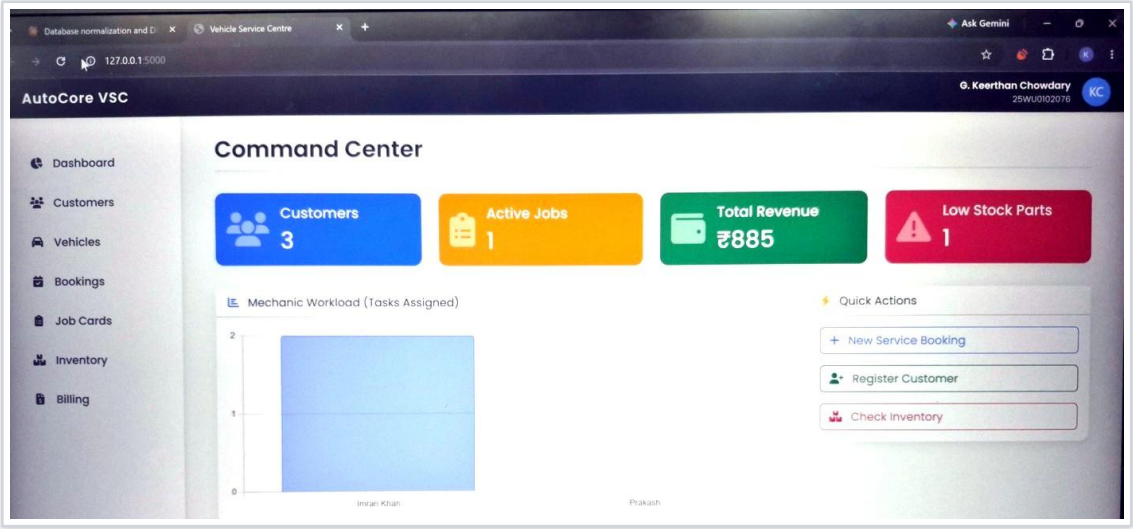
3

Review 3

Website

AutoCore VSC web interface, final demonstration, report and viva-voce

AutoCore VSC: The Website



AutoCore VSC

Web interface for the Vehicle Service Centre database, opened in a browser.

Where it runs

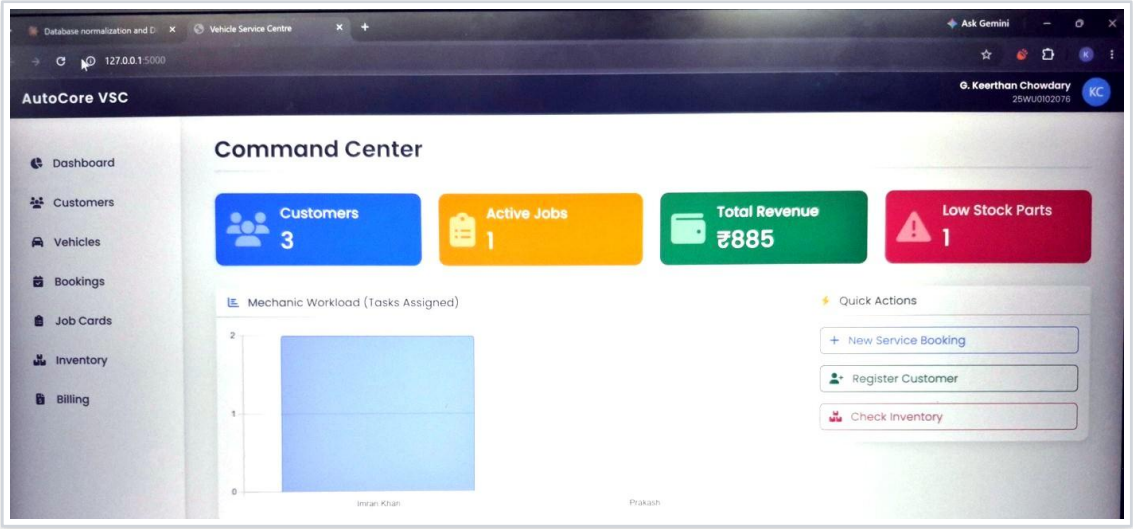
Locally at 127.0.0.1:5000 and shared online through the public link on the next line.

Seven pages

Dashboard, Customers, Vehicles, Bookings, Job Cards, Inventory, Billing.

https://89cfa03d00e924.lhr.life/

Dashboard: Command Center



Live summary

Four cards: Customers 3, Active Jobs 1, Total Revenue Rs. 885 and Low Stock Parts 1.

Mechanic workload

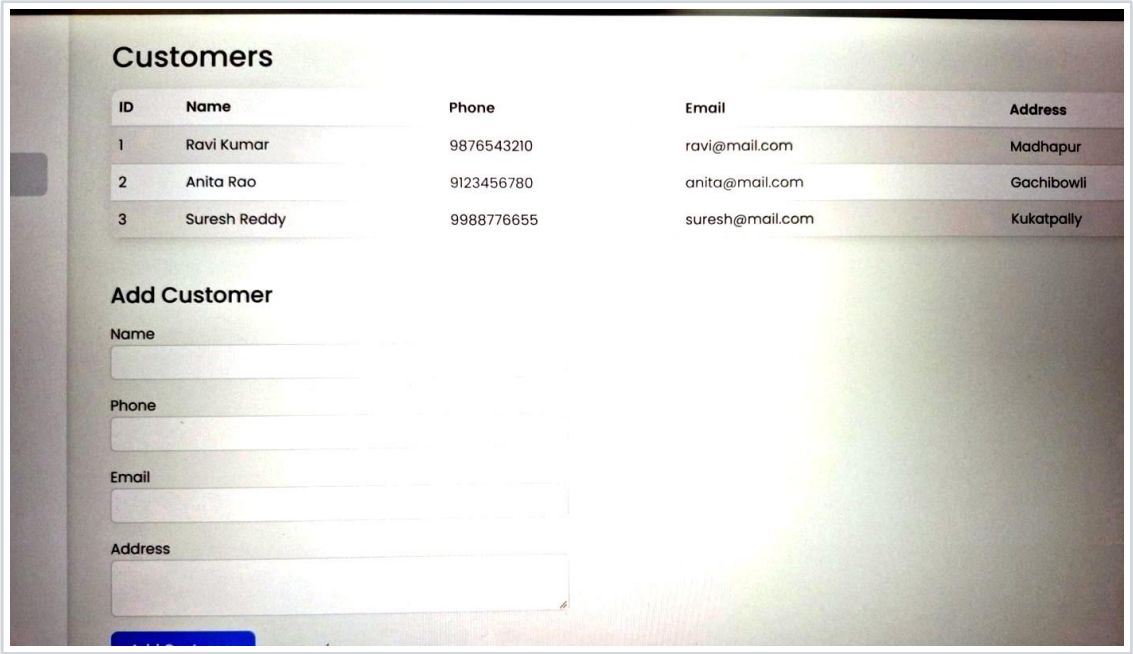
Bar chart of tasks assigned to each mechanic (Imran Khan, Prakash).

Quick actions

New Service Booking, Register Customer and Check Inventory in one click.

Screenshot of the AutoCore VSC website running at 127.0.0.1:5000

Customers Page



Customer list

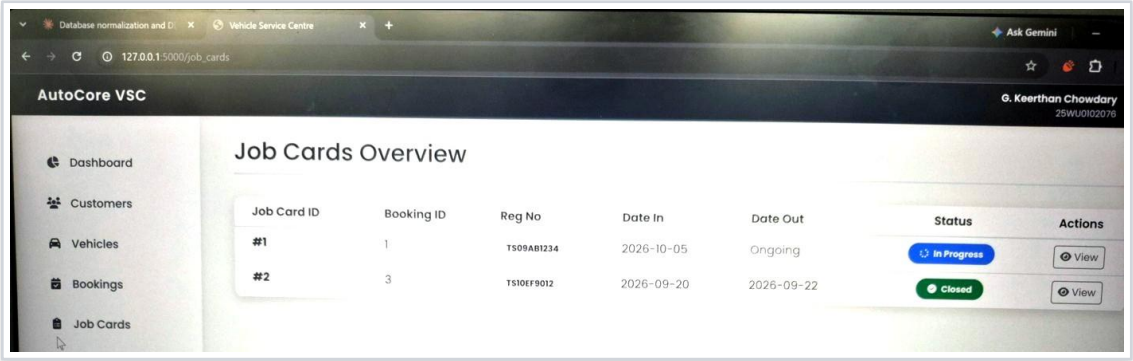
ID, name, phone, email and address of every registered customer.

Add Customer form

Name, phone, email and address saved to the database with one button.

Screenshot of the AutoCore VSC website running at 127.0.0.1:5000

Job Cards Overview



Screenshot of the AutoCore VSC website running at 127.0.0.1:5000

Job card table

Job card ID, booking ID, registration number, date in and date out.

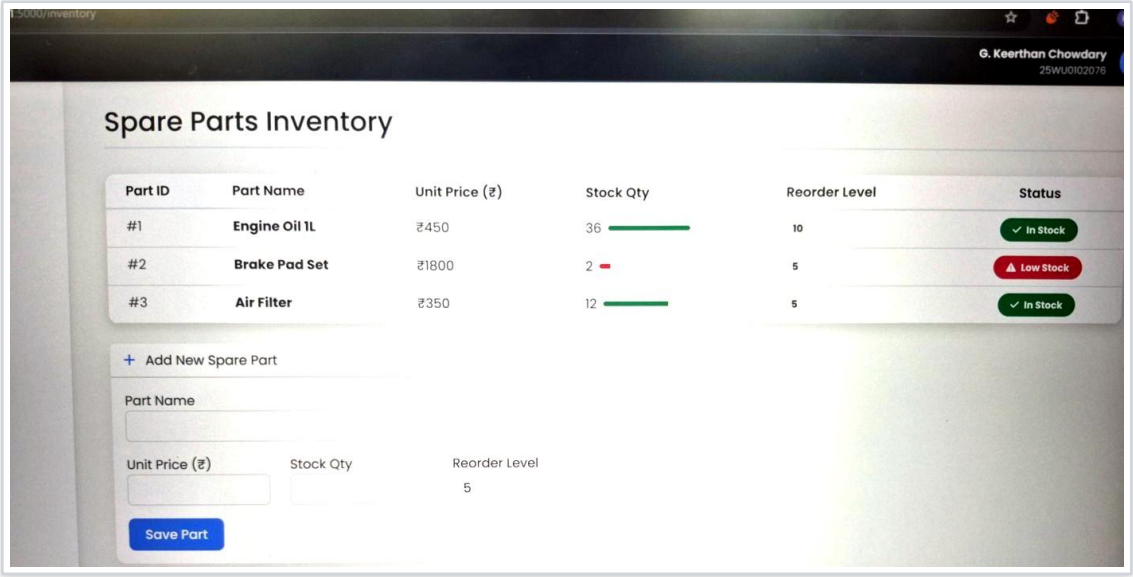
Status badges

In Progress for the open job (date out shows Ongoing) and Closed for the finished one.

View action

Each row has a View button to open the job card details.

Spare Parts Inventory



Stock at a glance

Part, unit price in Rs., stock quantity with a level bar, and the reorder level.

Low-stock warning

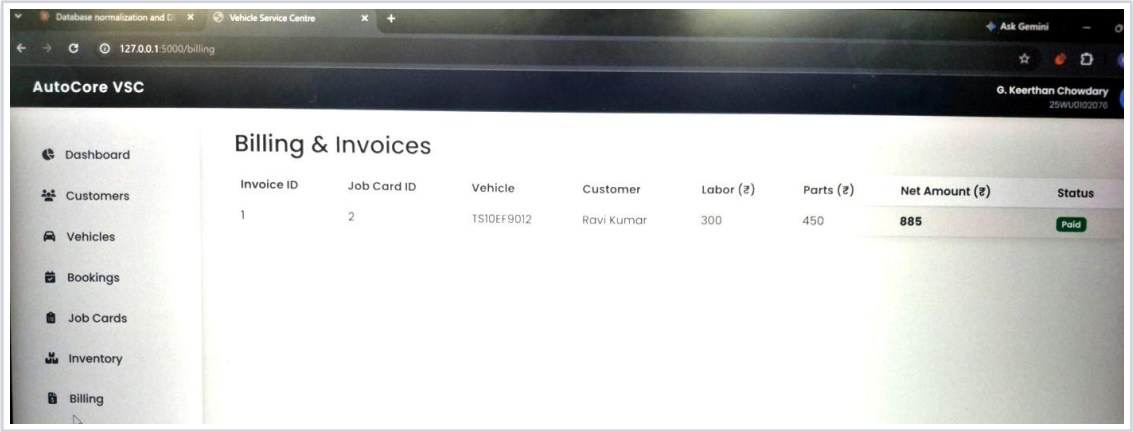
Brake Pad Set shows Low Stock (2 left, reorder level 5); the others show In Stock.

Add New Spare Part

Form for part name, unit price, stock quantity and reorder level (default 5).

Screenshot of the AutoCore VSC website running at 127.0.0.1:5000

Billing & Invoices



Invoice details

Invoice, job card, vehicle, customer, labor, parts and net amount.

Sample invoice

Invoice 1 for job card 2: labor Rs. 300 + parts Rs. 450 + 18% GST = Rs. 885.

Payment status

Shown as a Paid badge; this amount feeds the dashboard revenue.

Screenshot of the AutoCore VSC website running at 127.0.0.1:5000

Using the Website: End-to-End Workflow



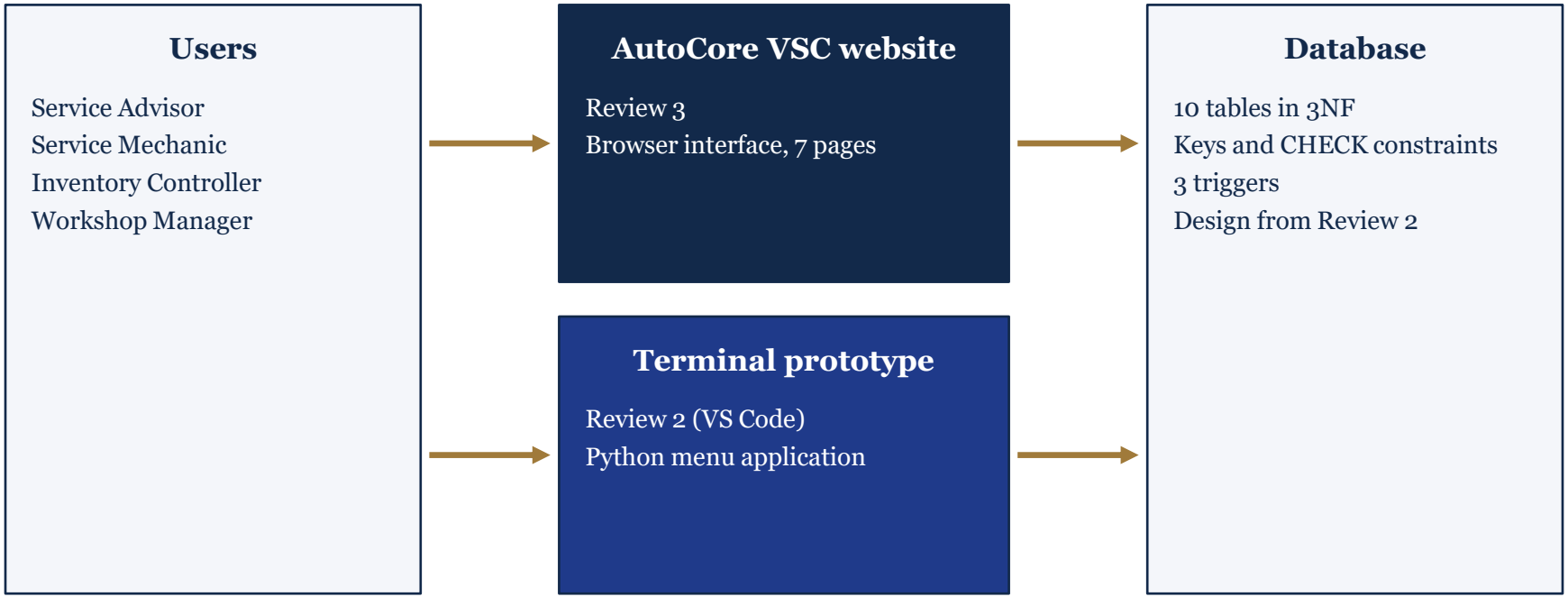
Sidebar navigation

Every step is one click away in the left sidebar, so a service advisor can follow a vehicle from booking to invoice.

Sample scenario

Customer Ravi Kumar, vehicles TS09AB1234 and TS10EF9012, job cards 1 and 2, invoice paid at Rs. 885.

Architecture: Website and Review 2 Prototype



Business Rules Enforced by the Database

10 Tables in 3NF

3 Triggers

4 Constraint types: PK/FK, UNIQUE, CHECK, NOT NULL

trg_no_overlap

Before an Assignment insert, rejects a mechanic booked in an overlapping time slot.

trg_check_stock

Before a Part_Usage insert, aborts with "Insufficient stock" if quantity exceeds stock.

trg_deduct_stock

After a Part_Usage insert, reduces Spare_Part.StockQty by the quantity issued.

Rules live in the database, so every route into the data obeys them.

Testing & Results

#	TEST CASE	EXPECTED BEHAVIOUR	RESULT
1	Duplicate booking, same vehicle and slot	Rejected by UNIQUE constraint	PASS
2	Overlapping task for the same mechanic	Blocked by trg_no_overlap	PASS
3	Issue 5 brake pad sets (stock is 3)	Blocked by trg_check_stock	PASS
4	Issue 1 brake pad set	Allowed, stock falls from 3 to 2	PASS
5	Invoice for job card 1	(1100 + 3600) x 1.18 = 5546.00	PASS
6	Low-stock report after usage	Brake Pad Set (2 left, level 5)	PASS

Tests were run on the database and Review 2 terminal prototype using the sample scenario.

Conclusion & Future Scope

Achievements

- Designed and normalized the database to 3NF (Reviews 1 and 2)
- Built and tested a terminal prototype in VS Code (Review 2)
- Built the AutoCore VSC website with dashboard, customers, vehicles, bookings, job cards, inventory and billing (Review 3)
- Business rules enforced inside the database

Future Scope

- Permanent hosting of the website
- Role-based login for the four user roles
- SMS or email service reminders
- Online payment integration
- Deployment on MySQL

Thank You

Questions & Viva-Voce

ER Modeling

3NF Normalization

Constraints & Triggers

SQL Queries

AutoCore VSC Website